CRICKET MATCH DATA ANALYSIS

Prepared by: Duraikannan S

1. Project Review

This project analyzes cricket match data using Python. The objective is to identify trends, team performance, and player insights using data visualization.

2. Dataset Description

The dataset includes match details such as teams, scores, venue, winner, top scorer, and match date. Data processing was done using Pandas.

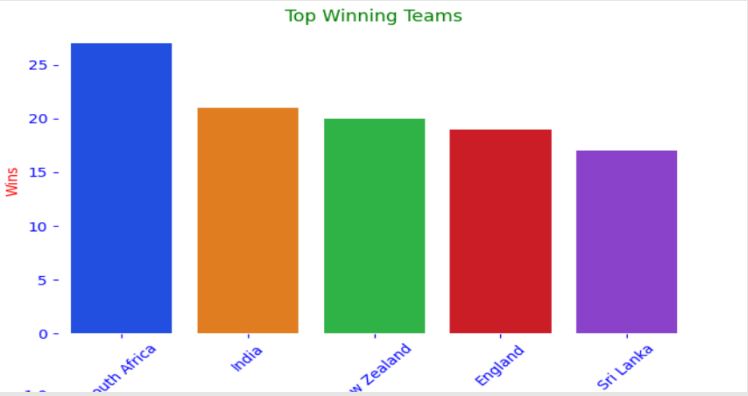
3. Analysis Explanation

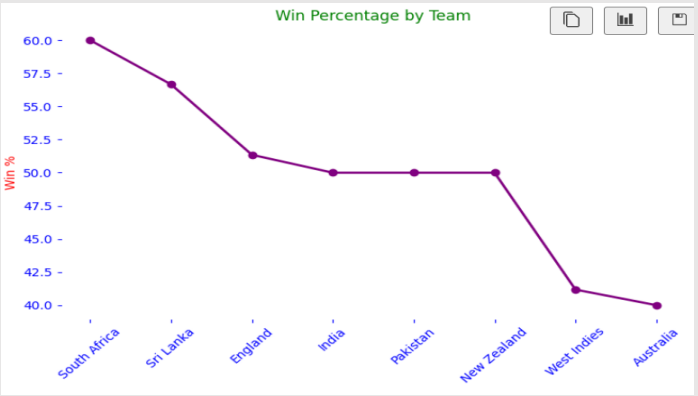
This analysis includes:

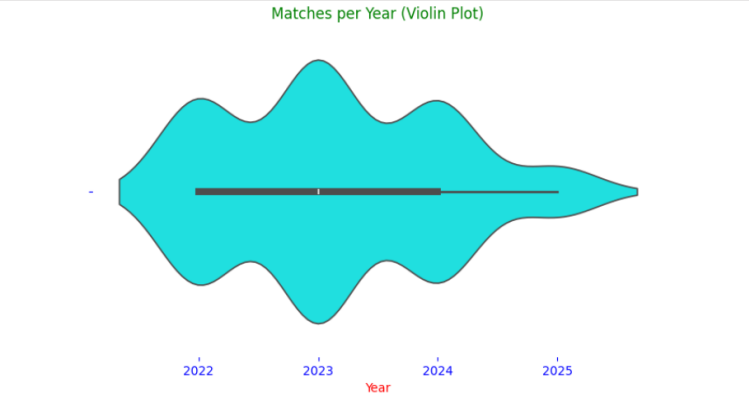
1. Top Winning Teams – shows which teams have the highest wins.
2. Win Percentage – compares team efficiency.
3. Matches per Year – shows match distribution over time.
4. Top Scorers – highlights most frequent top scorers.
5. Total Runs Distribution – shows scoring patterns.
6. Average Runs – compares team performance.
7. Top Venues – identifies popular match locations.
8. Man of the Match – shows top-performing players.

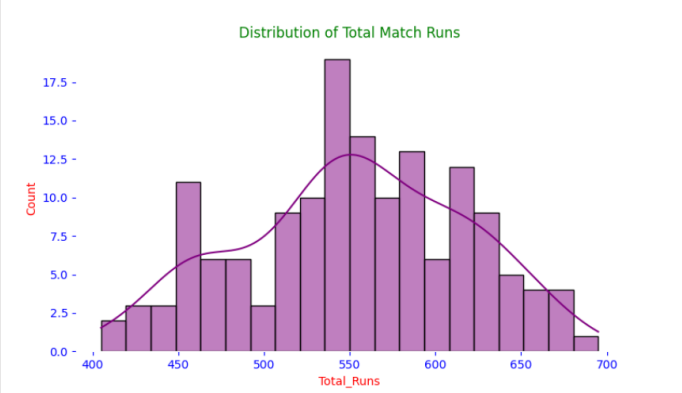
4. Key Insights

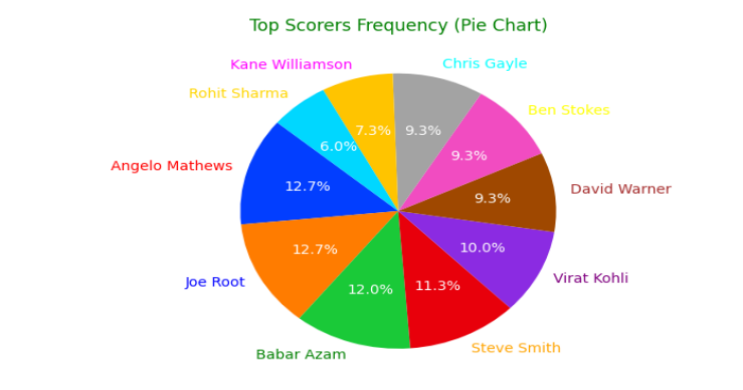
- Top teams dominate win statistics.
- Certain players frequently win Man of the Match.
- Match scores vary across venues and years.
- Performance trends show team consistency differences.

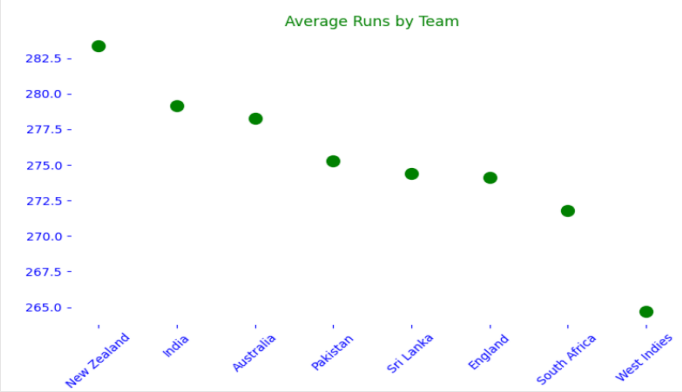


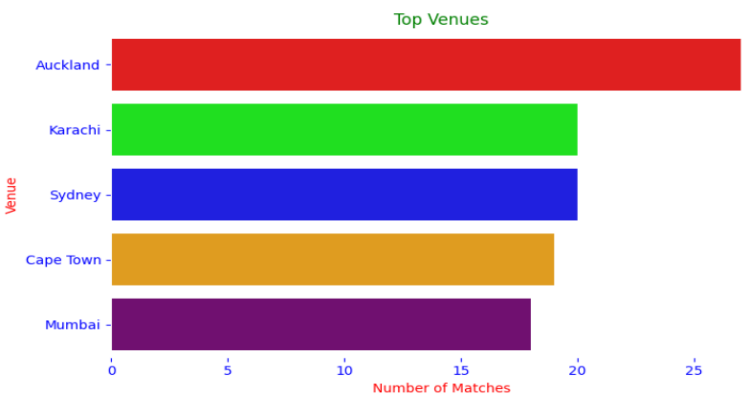


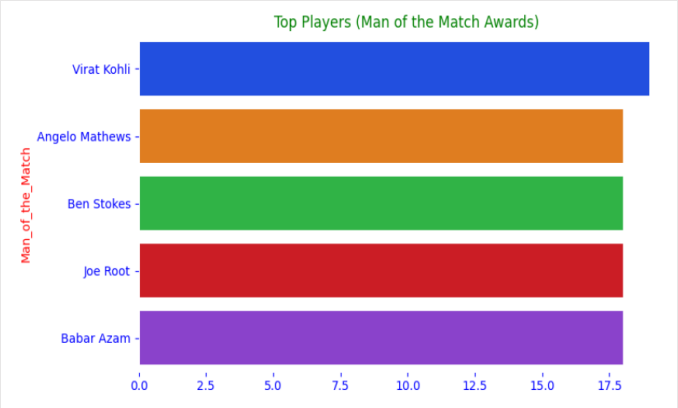












5. VS Code

```
# Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
df = pd.read_csv("cricket_matches.csv")

# Convert Date column
df['Date'] = pd.to_datetime(df['Date'])
df['Year'] = df['Date'].dt.year

# Create subplot grid (4 rows × 2 columns for 8 charts)
fig, axes = plt.subplots(4, 2, figsize=(16, 20))
axes = axes.flatten()

for ax in axes:
    # Spine colors
    for spine in ax.spines.values():
        spine.set_color('white')  # all borders white

    # Tick label colors
    ax.tick_params(axis='x', colors='blue')
    ax.tick_params(axis='y', colors='blue')

    # Axis label colors
    ax.xaxis.label.set_color('red')
    ax.yaxis.label.set_color('red')

    # Title color
    ax.title.set_color('green')

# 1. Top Winning Teams
top_teams = df['Winner'].value_counts().head(5)
sns.barplot(x=top_teams.index, y=top_teams.values, palette='bright', ax=axes[0])
axes[0].set_title("Top Winning Teams")
axes[0].set_ylabel("Wins")
```

```
axes[0].tick_params(axis='x', rotation=45)

# 2. Top Venues
venue_counts = df['Venue'].value_counts().head(5)
sns.barplot(x=venue_counts.values, y=venue_counts.index, palette=['#FF0000', '#00FF00', '#0000FF', '#FFA500',
'#800080'], ax=axes[6])
axes[6].set_title("Top Venues")
axes[6].set_xlabel("Number of Matches")

# 3. Top Players (Man of the Match)
mom_counts = df['Man_of_the_Match'].value_counts().head(5)
sns.barplot(x=mom_counts.values, y=mom_counts.index, palette='bright', ax=axes[7])
axes[7].set_title("Top Players (Man of the Match Awards)")

# 4. Win Percentage by Team
total_matches = df['Team1'].value_counts().add(df['Team2'].value_counts(), fill_value=0)
win_counts = df['Winner'].value_counts()
win_percentage = (win_counts / total_matches * 100).sort_values(ascending=False)
axes[1].plot(win_percentage.index, win_percentage.values, marker='o', color='purple', linewidth=2)
axes[1].set_title("Win Percentage by Team")
axes[1].tick_params(axis='x', rotation=45)
axes[1].set_ylabel("Win %")

# 5. Average Runs by Team
avg_scores = pd.concat([
    df.groupby('Team1')['Team1_Score'].mean(),
    df.groupby('Team2')['Team2_Score'].mean()
]).groupby(level=0).mean().sort_values(ascending=False)
axes[5].scatter(avg_scores.index, avg_scores.values, color='green', s=100)
axes[5].set_title("Average Runs by Team")
axes[5].tick_params(axis='x', rotation=45)

# 6. Distribution of Total Match Runs
df['Total_Runs'] = df['Team1_Score'] + df['Team2_Score']
```

```
sns.histplot(df['Total_Runs'], bins=20, kde=True, color='purple', ax=axes[4])
axes[4].set_title("Distribution of Total Match Runs")

# 7. Top Scorers Frequency (Pie Chart)
top_scorer_counts = df['Top_Scorer'].value_counts().head(10)

# Create pie chart and capture text objects
wedges, texts, autotexts = axes[3].pie(
    top_scorer_counts.values,
    labels=top_scorer_counts.index,
    autopct='%1.1f%%',
    startangle=140,
    colors=sns.color_palette('bright', len(top_scorer_counts))
)

# Custom label colors for each scorer
label_colors = ['red', 'blue', 'green', 'orange', 'purple',
                'brown', 'yellow', 'cyan', 'magenta', 'gold']

# Apply colors to scorer labels
for text, color in zip(texts, label_colors):
    text.set_color(color)

# Keep percentage labels white for contrast
for autotext in autotexts:
    autotext.set_color('white')

axes[3].set_title("Top Scorers Frequency (Pie Chart)")

# 8. Matches per Year (Violin Plot)
sns.violinplot(x=df['Year'], ax=axes[2], color="cyan")
axes[2].set_title("Matches per Year (Violin Plot)")
```

```
# Adjust spacing
plt.tight_layout()
plt.tight_layout()
plt show()
```

6. Conclusion:

This project analyzed cricket match data using python to find key insights and trend. It shows how data analysis helps in understanding team performance and match outcome..